

C A S E R E P O R T

Breast cancer seeding the needle tract after tru-cut needle biopsy (TCNB): Case report and literature review

FEDERICA CIGLIANO¹, ANNALISA GORGOGNONE², VINCENZA TALAMO²,
FILOMENA CINELLI², MARTINA PIA PAGLIARA², LUCA PAZIENZA²,
GIUSEPPE GUGLIELMI^{3,4}

¹Radiology Unit, "Policlinico Foggia" Hospital, Foggia, Italy; ²Radiology Unit, "Casa Sollievo della Sofferenza" Hospital, San Giovanni Rotondo, Foggia, Italy; ³Department of Clinical and Experimental Medicine, Foggia University School of Medicine, Foggia, Italy; ⁴Radiology Unit, "Dimiccoli" Hospital, Barletta, Italy

ABSTRACT

Needle tract seeding occurs when tumor cells are inadvertently planted along the path of a needle during biopsy. It's a possible occurrence related to breast needle biopsy. Although the clinical significance of such dissemination is statistically not significant, theoretical risk factor to develop recurrent locoregional disease, following needle biopsy of suspicious lesions, is recognized. This study reports a right breast cancer seeding the needle tract case, after Tru-Cut needle biopsy (TCNB). (www.actabiomedica.it)

Key words: breast cancer, needle tract seeding, tru-cut needle biopsy



Received: 10 February 2026 | Accepted: 13 March 2026

Correspondence: Giuseppe Guglielmi, MD / Department of Clinical and Experimental Medicine, Foggia University School of Medicine, Foggia, Italy. Radiology Unit, "Dimiccoli" Hospital, Viale Ippocrate 15, 76121, Barletta, Italy / E-mail: giuseppe.guglielmi@unifg.it / ORCID: 0000-0002-4325-8330

Introduction

Breast needle biopsy is the best technique for sampling fragments of suspicious breast lesions. Old studies already defined high levels of sensitivity and specificity of TCNB method: the sensitivity rate herein was 89%, the specificity rate was 100% and the over-all accuracy rate was 88% (1); the sensitivity rate was 89%, the specificity rate was 100% and the predictive value was 100% (2). A more recent study compares FNAC and CNB in the preoperative evaluation of suspicious breast masses; CNB has a sensitivity rate of 96,97%, a specificity rate of 100%, a PPV of 100%, a NPV of 91,67% and a diagnostic accuracy of 97,73% (3). The concordance level between core needle biopsy and surgical resection specimen for overall histologic grading was 73%: for tubule formation - 71%, for nuclear pleomorphism - 91%, for the mitotic index - 59% (4). Despite these data, needle biopsy is related with tumour cells dissemination at the site of needle passage during needle biopsy. However, the most important data is the possible correlation with an increased risk of local/regional recurrence over time. Scientific literature defines this risk as a theoretical risk that does not impact the long-term survival of patients (5-7).

Case description

In August 2024, a 45-year-old woman, came to our breast diagnostic department because of a palpable mammary right swelling. Therefore, mammographic and elastosonographic investigations were performed. Digital breast tomosynthesis and ultrasound detected a lesion in the right upper inner quadrant, with irregular shape, micro lobulated margins and size about 30 mm, highly suggestive of malignancy (BIRADS 5) (Figures 1-3). DBT and US also detected 23 mm lymph node in right axilla, suspicious for metastatic lymph node.

A US-guided needle biopsy was performed. As per current protocol in our department, 14-gauge automated cutting needle biopsy device was used for the breast nodule and 4 samples were taken; 18-gauge automated cutting needle biopsy device was used for the axillary lymph node and 2 samples were taken. Breast nodule biopsies were performed with access from the



Figure 1. Right breast ultrasound showing 30 mm hypoechoic mass lesion with irregular shape, microlobulated margins in the upper inner quadrant of the right breast.

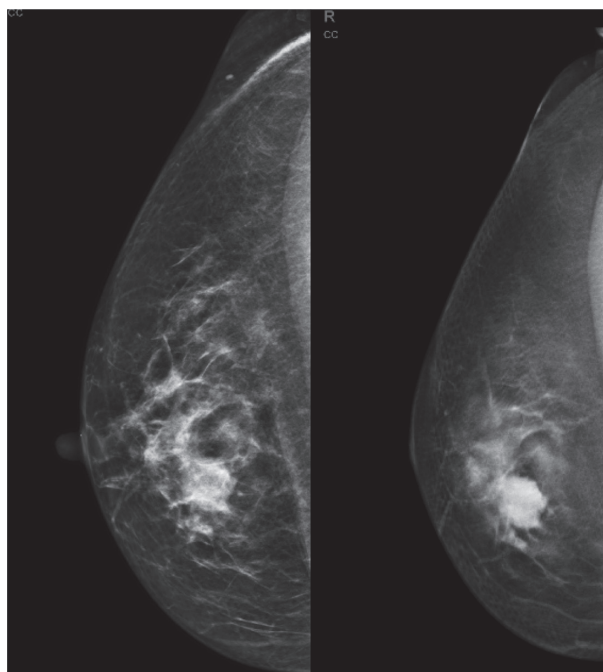


Figure 2. Right mammogram CC (2D and tomosynthesis) view showing hyperdense and lobulated mass lesion, with irregular shape in the upper inner quadrant of the right breast.

outer upper quadrant of the right breast. Histopathological diagnosis of the breast mass was invasive ductal carcinoma (IDC), triple negative, PDL-1 positive. Histopathological diagnosis of the axillary lymph node

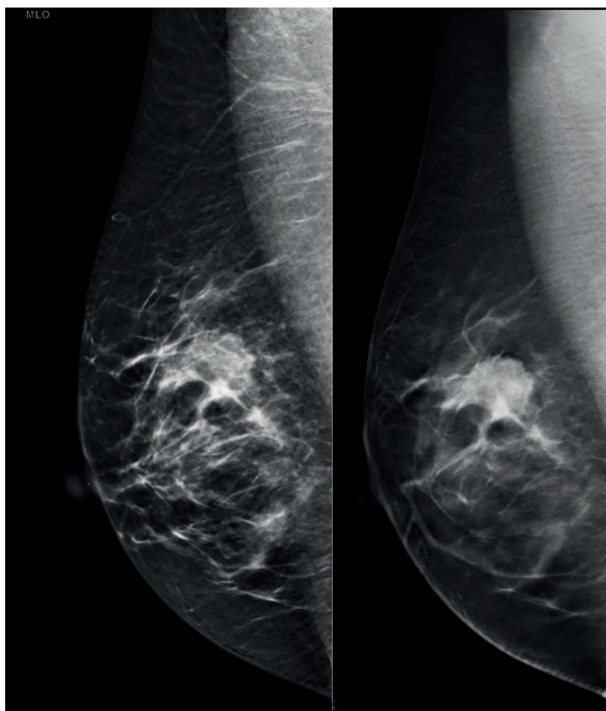


Figure 3. Right mammogram MLO (2D and tomosynthesis) view showing hyperdense and lobulated mass lesion, with irregular shape in the upper inner quadrant of the right breast.

was IDC breast carcinoma metastasis, triple negative. The same month, the patient was submitted to contrast enhancement total body CT scan. CT scan confirmed the right breast mass without metastasis. It also highlighted a linear fibrotic strand in the right breast adipose tissue, surrounded by fluid imbibition, between the mass and the skin (Figure 4).

This sign was compatible with the needle tract. Radiological staging was T2N1M0. In September 2024, the patient received neoadjuvant chemotherapy (carboplatin and anthracyclines). In December 2024 she was forced to interrupt it because of the onset of acute renal failure. In January 2025, 3 months after the diagnosis, she was reevaluated in our breast diagnostic department with severe worsening in the mammary site. Therefore, mammographic and elastosonographic investigations were performed. Digital breast tomosynthesis and ultrasound detected growing right breast mass. It appeared bulkier, stretched, size about 70 mm, extending from right upper inner quadrant to the skin. It caused thickened and retracted skin in that site

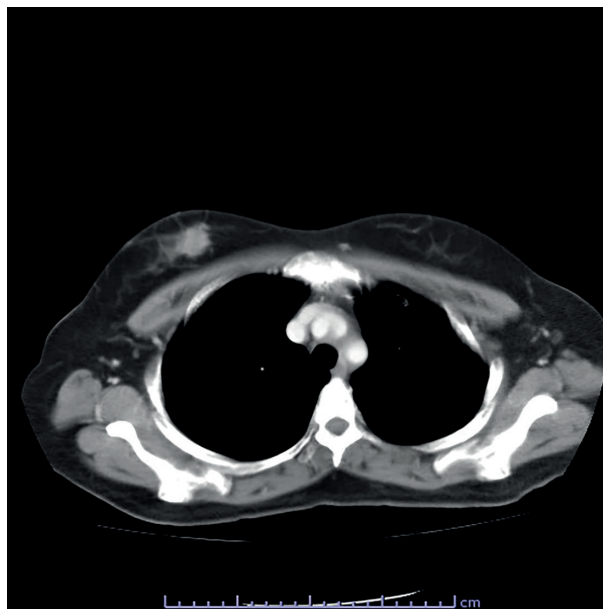


Figure 4. Contrast enhancement CT scan showing right breast mass and a linear fibrotic strand in the right breast adipose tissue, between the mass and the skin, compatible with the needle tract.

(Figures 5-7). The mass was placed along the needle tract. The right axillary lymph node was unchanged.

In February 2025, total body CT scan confirmed the presence of the bulky mass, measuring approximately 65x16 mm (Figure 8). There were no metastases. Radiological staging was T4bN1M0. Instrumental examination was compatible with regional spread of disease.

The same month, the patient was also treated by surgery; right nac-sparing mastectomy and right axillary lymphadenectomy were performed. The histopathological diagnosis of needle tract was invasive ductal carcinoma extended to the superficial dermis of the overlying skin.

Discussion

Needle biopsy is the standard procedure to obtain a histological diagnosis of a suspicious breast lesion. Its multiple advantages have been recognized for years: reducing healthcare costs and the psychological and aesthetic impact if compared to surgical excisional

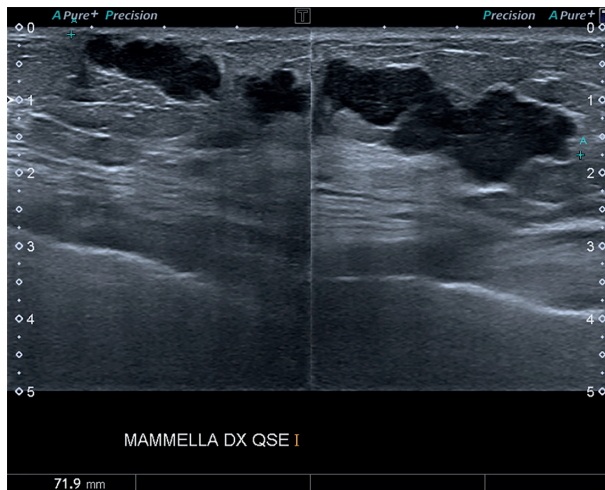


Figure 5. Right breast ultrasound showing 70 mm bulkier, stretched and hypoechoic mass lesion with irregular shape, microlobulated margins extending from the upper inner quadrant of the right breast to the skin.

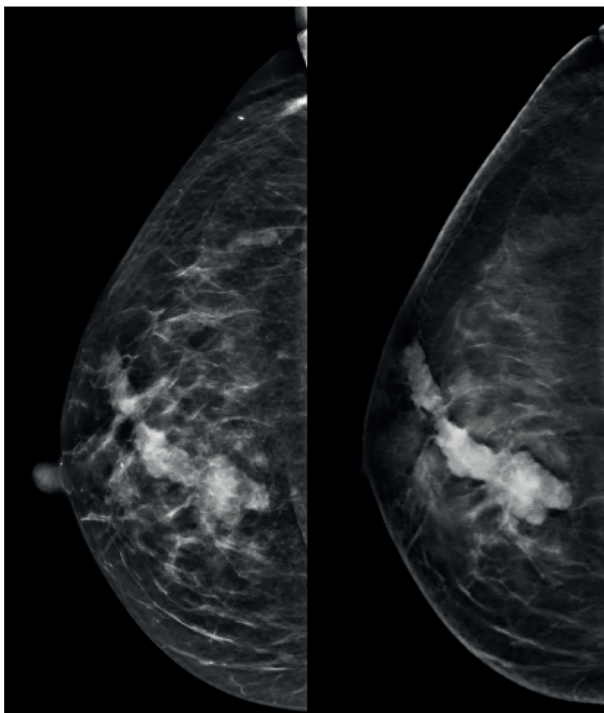


Figure 6. Right mammogram CC (2D and tomosynthesis) view showing bulkier, stretched and hyperdense mass lesion with irregular shape, microlobulated margins extending from the upper inner quadrant of the right breast to the skin.

biopsies (8, 9), defining the most appropriate surgical approach (10-12), evaluating the usefulness of neoadjuvant chemotherapy by evaluating molecular phenotype (13). An event related to breast needle biopsies

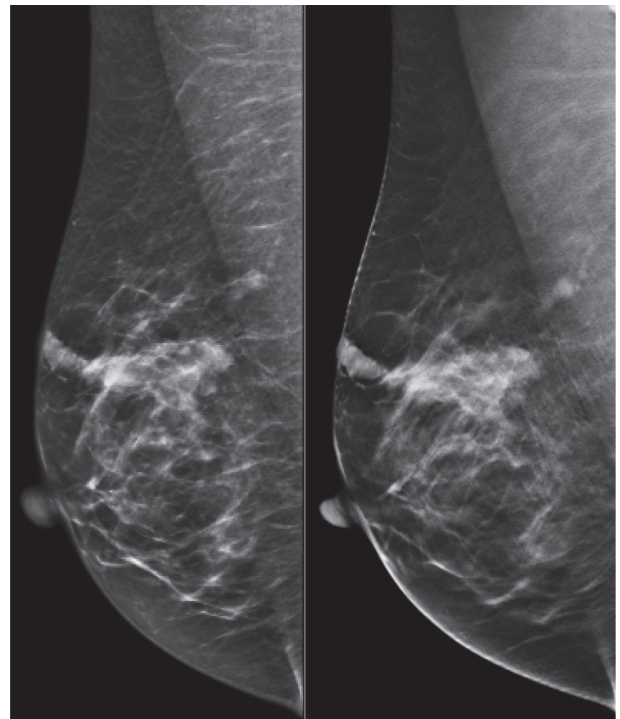


Figure 7. Right mammogram MLO (2D and tomosynthesis) view showing bulkier, stretched and hyperdense mass lesion with irregular shape, microlobulated margins extending from the upper inner quadrant of the right breast to the skin.

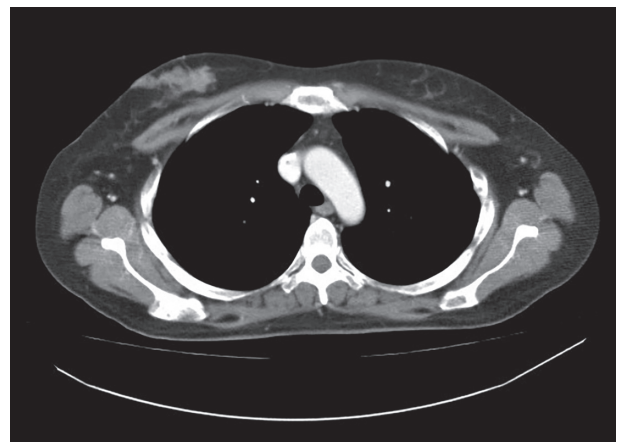


Figure 8. Contrast enhancement CT scan showing the right breast bulky mass extending from the upper inner quadrant of the right breast to the skin.

is needle tract seeding of viable tumour cells. The incidence of this phenomenon is difficult to evaluate as it's highly variable, perhaps because of different needle biopsy devices used (CNB vs FNAB) (14), number of samples taken (75% in multiple CNB passes vs 66%

in single CNB pass) (15), distance of time between biopsy and surgery (6,16,17), tumour histology (18). The post-CNB incidence assessed on histopathological examination of the surgical specimens is 22-50% (19); the incidence on cytology deriving from the needle washing material after CNB is 65% (15). Other studies, however, define the risk of needle tract seeding tumour cells extremely low (0-2%) (6,16,17,20,21,22,23). The possibility of finding viable malignant cells along the needle tract is inversely proportional to the time between biopsy and surgical excision (16), both due to the apoptosis of tumour cells and to natural immune elimination (6,17). According to the study authors even the histology of the tissues has an impact on the rate of dissemination of viable neoplastic cells with a percentage of 40% in IDC, 15% in infiltrating lobular carcinoma and 16% in mixed infiltrating ductal and lobular carcinoma. The most interesting feature of the present study is the clinical correlation between needle tract seeding of viable neoplastic cells and loco-regional spread or recurrence of the disease. In this circumstance literature data show a greater degree of concordance, so that the clinical significance of such dissemination is statistically insignificant (15). Moreover, the dissemination of viable neoplastic cells after biopsy has no significant impact on the rate of local recurrence of breast cancer, distant metastasis and long-term survival (5-7). The dissemination of viable neoplastic cells may increase the probability of local recurrence but has no impact on regional recurrence and long-term survival (18). Despite the low statistical significance of this phenomenon, many preventive procedures have been advanced to reduce the risk of tumour cells seeding and the related dissemination of disease. The administration of epinephrine together with the local anaesthetic solution, the use of an introducer or coaxial and the use of cryoablation-assisted needle biopsy represent preventive measures of dissemination (18). Another preventive measure of dissemination is the use of radiofrequency pulses to sterilize the needle tract and denature viable tumour cells (24). Postoperatively, radiotherapy is recommended for both in situ and invasive carcinoma to sterilize the needle tract, if it is not included in the surgical specimen (16). This case report is an example of rare regional spread of disease after US-guided TCNB of a right breast malignant nodule. In accordance with the relevant literature, our patient

was affected by infiltrating breast carcinoma, triple negative molecular subtype. The triple negative molecular subtype PDL-1 positive and the metastatic disease required neoadjuvant chemotherapy based on carboplatin and anthracyclines, early interrupted due to the onset of adverse effects (acute renal failure). From the interruption of chemotherapy in December 2024 until February 2025, the patient showed rapid disease progression. Imaging allowed to advance the hypothesis of needle tract spread of disease. In January and February 2025 instrumental investigations (mammography, US and total body CT scan) showed the solid elongated mass, extending from right breast upper internal quadrant (first location of disease) towards the upper-external quadrants (needle passage site) and up to the skin (needle access site). Nodule distribution follows the entire needle tract performed during US-guided TCNB in August 2024. It is also documented by mammography, US and total body CT scan performed the same month. The regional spread of disease was expected to be continuous. The gold standard to confirm our diagnostic hypothesis is post-operative histopathological needle tract examination. The needle tract was included in the mastectomy, and it was analysed by the pathologist. In accordance with the histotype obtained by TCNB performed in August 2024, the histological examination of the needle tract demonstrated in the right breast outer upper quadrant a neoplastic nodule of approximately 4 cm corresponding histologically to infiltrating ductal carcinoma, extending up to the superficial dermis of the overlying skin. Resection margins are clean. This state confirms the imaging suspicion of regional spread of disease after needle tract seeding of tumour cells. Histopathological examination of axillary lymph nodes and axillary adipose tissue shows 18 lymph nodes with metastatic disease and the presence of viable neoplastic cells in the peri-lymph node fat.

Ethic Approval: Radiology Unit, "Casa Sollievo della Sofferenza" Hospital, San Giovanni Rotondo, Foggia, Italy
EMOST 01/2011 – V 1_30 Nov 11.

Conflict of Interest: Each author declares that he or she has no commercial associations (e.g. consultancies, stock ownership, equity interest, patent/licensing arrangement etc.) that might pose a conflict of interest in connection with the submitted article.

Authors' Contribution: FC (first author), AG (senior author), VT (intermediate co-author, data acquisition and interpretation of results), FC (intermediate co-author, data acquisition and interpretation of results), MPP (intermediate co-author, data acquisition and interpretation of results), LP (intermediate co-author, data acquisition and interpretation of results), GG (corresponding author).

Declaration on the Use of AI: None.

Consent for Publication: Informed consent was obtained from all subjects involved in this study.

Funding: None.

References

1. Cusick JD, Dotan J, Jaecks RD, Boyle Jr WT. The role of Tru-Cut needle biopsy in the diagnosis of carcinoma of the breast. *Surg Gynecol Obstet.* 1990;170(5):407-10.
2. Minkowitz S, Moskowitz R, Khafif RA, Alderete MN. TRU-CUT needle biopsy of the breast. An analysis of its specificity and sensitivity. *Cancer.* 1986 Jan 15;57(2):320-3. doi:10.1002/1097-0142(19860115)57:2<320::aid-cncr2820570221>3.0.co;2-c.
3. Pradhan M, Mandal A, Biswas BK, Hazra A, Kumar D. The Role of Fine-Needle Aspiration Cytology in the Evaluation of Breast Lumps. *Cureus.* 2024 Oct 3;16(10):e70748. doi: 10.7759/cureus.70748. eCollection 2024 Oct.
4. Szpor J, Witczak K, Storman M, et al. Breast carcinoma grading on core needle biopsy - to grade or not to grade? *Pol J Pathol.* 2023;74(3):203-210. doi: 10.5114/pjp.2023.132225.
5. Sopik V, Narod SA. The relationship between tumour size, nodal status and distant metastases: on the origins of breast cancer. *Breast Cancer Res Treat.* 2018;170(3):647-656. doi: 10.1007/s10549-018-4796-9.
6. Diaz LK, Wiley EL, Venta LA. Are malignant cells displaced by large-gauge needle core biopsy of the breast? *AJR Am J Roentgenol.* 1999;173(5):1303-13. doi: 10.2214/ajr.173.5.10541110.
7. Schirrmacher V. Cancer metastasis: experimental approaches, theoretical concepts, and impacts for treatment strategies. *Adv Cancer Res.* 1985;43:1-73. doi: 10.1016/s0065-230x(08)60942-2.
8. Bruening W, Fontanarosa J, Tipton K, Treadwell JR, Launders J, Schoelles K. Systematic review: comparative effectiveness of core-needle and open surgical biopsy to diagnose breast lesions. *Ann Intern Med.* 2010 Feb 16;152(4):238-46. doi:10.7326/0003-4819-152-1-201001050-00190.
9. Gutwein LG, Ang DN, Liu H, et al. Utilization of minimally invasive breast biopsy for the evaluation of suspicious breast lesions. *Am J Surg.* 2011 Aug;202(2):127-32. doi: 10.1016/j.amjsurg.2010.09.005.
10. National Accreditation Program for Breast Centers. Optimal Resources for Breast Cancer. 2024 Standards. NAPBC Guideline. Available online: https://accreditation.facs.org/accreditationdocuments/NAPBC/Standards/Optimal_Resources_for_Breast_Care_2024.pdf.
11. American Society of Breast Surgeons. Consensus Guideline on Image-Guided Percutaneous Biopsy of Palpable and Nonpalpable Breast Lesions. Available online: <https://www.breastsurgeons.org/docs/statements/Consensus-Guideline-on-Image-GuidedPercutaneous-Biopsy-of-Palpable-and-Nonpalpable-Breast-Lesions.pdf>.
12. Five Things Physicians and Patients Should Question. Commission on Cancer Choosing Wisely: An Initiative of the ABMI Foundation. Available online: <https://www.facs.org/media/wegncq3f/coclist.pdf>.
13. Holmes D, Colfry A, Czerniecki B, et al. Performance and Practice Guideline for the Use of Neoadjuvant Systemic Therapy in the Management of Breast Cancer. *Ann Surg Oncol.* 2015 Oct;22(10):3184-90. doi: 10.1245/s10434-015-4753-3.
14. Sennerstam RB, Franzén BSH, Wiksell HOT, Auer GU. Core-needle biopsy of breast cancer is associated with a higher rate of distant metastases 5 to 15 years after diagnosis than FNA biopsy. *Cancer Cytopathol.* 2017 Oct;125(10):748-756. doi: 10.1002/cncy.21909.
15. Uematsu T, Kasami M. Risk of needle tract seeding of breast cancer: cytological results derived from core wash material. *Breast Cancer Res Treat.* 2008 Jul;110(1):51-5. doi: 10.1007/s10549-007-9685-6.
16. Hoorntje LE, Schipper MEI, Kaya A, Verkooijen HM, Klinkenbijn JG, Borel Rinkes IHM. Tumour cell displacement after 14G breast biopsy. *Eur J Surg Oncol.* 2004 Jun;30(5):520-5. doi: 10.1016/j.ejso.2004.03.001.
17. Uematsu T, Kasami M. Risk of needle tract seeding of breast cancer: cytological results derived from core wash material. *Breast Cancer Res Treat.* 2008 Jul;110(1):51-5. doi: 10.1007/s10549-007-9685-6.
18. Holmes DR. Reducing the Risk of Needle Tract Seeding or Tumor Cell Dissemination during Needle Biopsy Procedures. *Cancers (Basel).* 2024 Jan 11;16(2):317. doi: 10.3390/cancers16020317.
19. Yoneyama K, Nakagawa M, Hara A. Local recurrence of breast cancer caused by core needle biopsy: Case report and review of the literature. *Int J Surg Case Rep.* 2020;72:318-321. doi: 10.1016/j.ijscr.2020.06.013.
20. Youngson BJ, Cranor M, Rosen PP. Epithelial displacement in surgical breast specimens following needling procedures. *Am J Surg Pathol.* 1994 Sep;18(9):896-903. doi: 10.1097/00000478-199409000-00004.
21. Stoller A, Skinner J, Levine EA. A prospective study of seeding of the skin after core biopsy of the breast. *Am J Surg.* 2000 Aug;180(2):104-7. doi: 10.1016/s0002-9610(00)00425-6.

22. Liberman L, Vuolo M, Dershaw DD, et al. Epithelial displacement after stereotactic 11-gauge directional vacuum-assisted breast biopsy. *AJR Am J Roentgenol.* 1999 Mar;172(3):677-81. doi: 10.2214/ajr.172.3.10063859.
23. Michalopoulos NV, Zagouri F, Sergeantanis TN, et al. Needle Tract Seeding after Vacuum-Assisted Breast Biopsy. *Acta Radiologica* 2008 May; 49(3):267-70. doi:10.1080/02841850701775030.
24. Wiksell H, Schässburger KU, Janicijevic M, et al. Prevention of tumour cell dissemination in diagnostic needle procedures. *Br J Cancer.* 2010 Nov 23;103(11):1706-9. doi: 10.1038/sj.bjc.6605964.

Copyright: The Author(s), 2026. Licensee Mattioli 1885, Fidenza, Italy. This is an open-access article distributed under the terms of the Creative Commons Attribution NonCommercial License (CC BY-NC-4.0).

Disclaimer/Publisher's Note: The statements, opinions and data contained in this article are solely those of the author(s) and contributor(s) and do not necessarily reflect those of their affiliated organizations, the publisher, the editors or the reviewers. The publisher and the editors disclaim any responsibility for injury to people or property resulting from any ideas, methods, instructions or products mentioned in the content. Any product that may be evaluated in this article, or claim made by its manufacturer, is not guaranteed or endorsed by the publisher.